

Harness Planning Document — Foreman

A harness that supervises a swappable AI worker doing **job-application review**. The worker only *proposes* an evaluation; the harness normalizes input, enforces declared guardrails, gates every output through pass/fail checkpoints, raises structured alarms, and persists each step — so constraint-handling is invisible to the agent.

DOMAIN: HIRING · HOURLY SCHOOL ROLES (CUSTODIAN / CAFETERIA) | AUTHOR: GONZALJ3 | EVENT: FIRED FESTIVAL · 24-HOUR BUILD CHALLENGE | AGENT = MODEL + HARNESS



THE FOUR PILLARS — DISTINCT MODULES, SEPARATE FROM THE WORKER

- ① **Material handling** **CLEAN I/O** — heterogeneous applications (PDF / scan / email) normalized to one canonical schema in; ranked decisions + audit trail out. The worker never touches raw files or APIs.
- ② **Guardrails** **DECLARED** — config-declared limits the worker can't bypass: no protected-class reasoning; every score must cite source evidence; **no qualification credited that isn't in the document**; hard-requirement gate (cert, background-check eligibility, work auth, availability); PII redaction. *e.g. worker writes "young, energetic" → blocked before output ships.*
- ③ **Checkpoints** **PASS / FAIL · PERSISTED** — explicit gates per applicant: schema valid · every criterion scored with citation · calibrated (not all-7s) · hard-gate computed. Results persisted → replay from any checkpoint.
- ④ **Alarms** **STRUCTURED** — named type + severity + context + recommended action; monitor the whole loop and drive escalation (e.g. **HALLUCINATED_QUALIFICATION** → critical → block + human review).

THE LOOP (WHAT'S GRADED)

- **Per-applicant correction loop:** worker proposes → guardrails + checkpoint gate → on fail the **specific failure is returned** and the worker **revises that one applicant** until it passes or escalates. This is the MUST: behavior changes on feedback.
- **Score absolutely, rank deterministically:** each applicant evaluated independently; the **harness** (not the LLM) sorts — pairwise tie-break only for the borderline cluster.
- **Calibration:** the 10 gold examples run as a regression set; drift fires an alarm.

SHOULD / BONUS

- **Swappable agent** interface — zero agent-specific code in the harness; **swap a 2nd model live** (bonus).
- **Persist & replay** from any checkpoint forward.
- **Human-in-the-loop** escalation on block / severity ≥ high.

STACK & DEPLOY

- **Python** harness core + **TypeScript** dashboard (live checkpoints / alarms); deployed URL for demo.

Out of scope (future): recruitment-marketing / ad-spend loop with budget arbitration — noted in HARNESS.md to keep the build tight.

Foreman · v0.1